

Total Marks – 84

Attempt Questions 1-7

All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your name and question number at the top of the page. Extra paper is available.

QUESTION 1 (12 MARKS) Begin a NEW sheet of writing paper. Marks

a) Solve $x - \frac{1}{x} < 0$ 2

b) For what values of x is $(2-x)(2x-1)(x+3) \leq 0$? 2

c) A committee of 3 has to be chosen from 4 males and 5 females. The committee must have at least 1 male and 1 female. How many different committees can be chosen? 2

d) Find $\int_0^1 x(2x-1)^4 dx$ using the substitution $u = 2x-1$ 3

e) The equation $x^3 + 2x^2 - 3x + 5 = 0$ has the roots α , β and γ 2

i) Find $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$, and $\alpha\beta\gamma$ 2

ii) Hence find the value of $(\alpha-1)(\beta-1)(\gamma-1)$ 1

QUESTION 2 (12 MARKS) Begin a NEW sheet of writing paper. Marks

a) Find $\frac{d}{dx} \ln \left(\frac{2x}{(x-1)^2} \right)$ 1

b) The gradient function of a certain curve is $(x^2 + 25)^{-1}$. Find the equation of the curve if it passes through the point $\left(5, \frac{\pi}{2}\right)$. 2

c) i) Sketch the curve $y = 3 \sin^{-1} 2x$. State its domain and range. 1

ii) Find the exact area bounded by the curve $y = 3 \sin^{-1} 2x$, the x axis and the line $x = \frac{1}{2}$. 3

d) Evaluate $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}}$ 2

e) Find the quotient $Q(x)$ and remainder $R(x)$ when the polynomial 3

$P(x) = 2x^4 - 3x^3 - x^2 + 2x + 1$ is divided by $x^2 + 2x - 1$.

QUESTION 3 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

- a) $\log_e x + \sin x = 0$ has a root close to $x = 0.5$. Using one application of Newton's method, find a better approximation to the root

2

b) $\int \cos^2\left(x - \frac{\pi}{4}\right) dx$

2

- c) Find the coefficient of x^9 in $\left(5x^2 - \frac{1}{2x}\right)^{12}$

2

- d) A particle is moving with acceleration $\ddot{x} = -9x$ and is initially stationary at $x = 4$.

i) Find v^2 as a function of x .

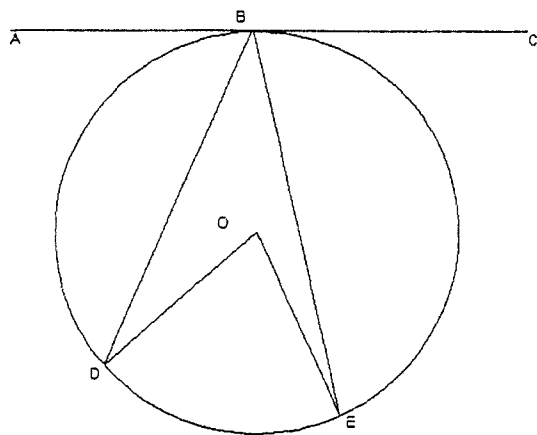
2

ii) What is the particle's maximum speed?

1

- e) In the diagram below, O is the centre of the circle, AC is a tangent at B and D and E are points on the circumference. If $\angle ABD = 80^\circ$ and $\angle DBE = 40^\circ$, find the size of $\angle BEO$, giving reasons.

3



QUESTION 4 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

- a) i) Prove the ratio of the $(k+1)$ th term to the k th term in the expansion of $\left(2x + \frac{3}{x}\right)^{12}$ simplifies to $\frac{39-3k}{2k}$

2

ii) Hence find the greatest coefficient of the expansion.

2

- b) i) Express $\sqrt{3} \cos 2t - \sin 2t$ in the form $R \cos(2t + \alpha)$ where $0 < \alpha < \frac{\pi}{2}$

2

ii) Hence or otherwise find all positive solutions of $\sqrt{3} \cos 2t - \sin 2t = 0$

2

- c) A particle moves in a straight line and is x metres from a fixed point O after t seconds where $x = 5 + \sqrt{3} \cos 2t - \sin 2t$.

i) Prove that the acceleration of the particle is $-4(x - 5)$.

2

ii) Between which two points does the particle oscillate? You may use your answers from part (b)

1

iii) When does the particle first pass through the point $x = 5$?

1

QUESTION 5 (12 MARKS) Begin a NEW sheet of writing paper.

Marks

a) Andy Roddick estimates that his chances of beating Rodger Federer are $\frac{1}{3}$

i) If 5 matches are played what is the probability Andy has exactly 3 wins.

1

ii) How many matches must Andy play so that the probability he wins at least one match is greater than 0.9?

2

b) A is the point (-4, 1) and B is the point (2, 4). Q is the point which divides AB internally in the ratio 2:1 and R divides AB externally in the ratio 2:1. P(x, y) is a variable point which moves so that PA = 2PB

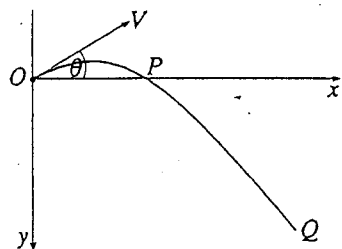
(i) Find the co-ordinates of Q and R.

2

(ii) Show that the locus of P is a circle with QR as diameter.

2

c)



A projectile is fired from O with speed $V \text{ ms}^{-1}$ at an angle of elevation of θ to the horizontal. After t seconds, its horizontal and vertical displacements from O as shown are x m and y m.

i) Given that $\ddot{x} = 0$ and $\ddot{y} = -g$, prove the equations of motion are $x = Vt \cos \theta$ and $y = -\frac{1}{2}gt^2 + Vt \sin \theta$.

2

ii) Find the time taken to reach P

1

iii) The projectile falls to Q where its angle of depression from O is θ . Prove that in its flight from O to Q that P is the half way point in terms of time.

2

Marks

QUESTION 6 (12 MARKS) Begin a NEW sheet of writing paper.

a) Use the identity $(1+x)^8(1+x)^8 = (1+x)^{16}$ to show:

$$\binom{8}{0}^2 + \binom{8}{1}^2 + \binom{8}{2}^2 + \dots + \binom{8}{8}^2 = \binom{16}{8}$$

3

b) Prove, by Mathematical induction, that for all positive integral values of n ,

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$$

4

c) i) Draw a sketch of $y = x$ and $y = \frac{3-x^2}{2}$ marking the coordinates of the points of intersection.

2

ii) Given $f(x) = \frac{3-x^2}{2}$ find the largest possible domain such that this function has an inverse.

1

iii) State the domain and range of the inverse function.

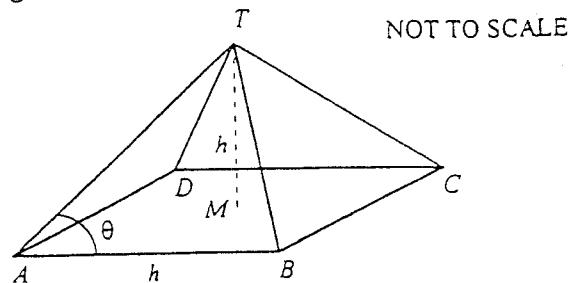
1

iv) Sketch the inverse function also on the same diagram

1

QUESTION 7 (12 MARKS) Begin a NEW sheet of writing paper. Marks

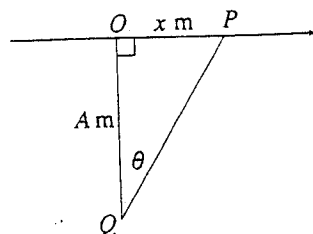
a) The diagram shows a right square pyramid with base ABCD, vertex T and altitude TM. It is given that $TM=AB=h$ units.



Show that if $\angle TAB = \theta^\circ$ then $\cos \theta^\circ = \frac{1}{\sqrt{6}}$

3

b) P is a point oscillating in simple harmonic motion on an x axis, the centre of motion being the origin, 0. The amplitude of the motion is A m, the period 2π seconds, and when $t=0$, the point is at 0 moving in the positive direction



i) Express x as a sine function of t .

1

ii) OQ is perpendicular to the axis, $OQ=A$ and $\angle OQP = \theta$
Show that $x = A \tan \theta$ and deduce that $\frac{dx}{d\theta} = A(1 + \sin^2 t)$

2

iii) Find $\frac{d\theta}{dt}$ as a function of t .

2

iv) Find the first time at which θ is increasing at a rate of $\frac{2}{7}$ radians/sec

1

c) The parabola $y^2 = x$ and $x^2 = 8y$ intersect at the origin and the point (a, b)

i) Find the values of a and b

1

ii) Prove that the curves divide the rectangle whose vertices are $(0,0)$ $(a,0)$ (a,b) $(0,b)$ into three regions of equal area.

2

QUESTION 1.

EXT 1 TRIAL 2005 SOLUTIONS

a) $x \neq 0$ critical point

$$x^2 - 1 = 0$$

$$(x-1)(x+1) = 0$$

$$\text{test } x = 2$$

$$x - \frac{1}{x} < 0 \text{ false}$$

$$\text{test } x = -\frac{1}{2}$$

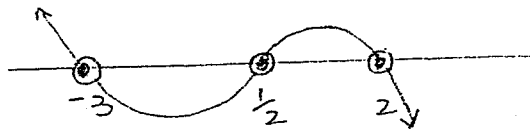
$$-\frac{1}{2} + 2 < 0 \text{ false}$$

Award 1
 $x \neq 0$ and
 $-1 < x < 1$
and no errors

[2]

$$\therefore x < -1 \text{ or } 0 < x < 1$$

b)



$$x = -3$$

$$x = \frac{1}{2}$$

$$x = 2$$

[2]

$$-3 \leq x \leq \frac{1}{2} \text{ and } x \geq 2$$

c) 1M 2F

$${}^4C_1 \times {}^5C_2 = 40$$

or

1F 2M

$${}^4C_2 \times {}^5C_1 = 30$$

$$\text{total} = 70$$

[2]

d)

$$\int_{-1}^1 u^4 \cdot \left(\frac{u+1}{2}\right) \cdot \frac{du}{2}$$

$$u = 2x - 1$$

$$\frac{du}{dx} = 2$$

$$du = 2dx$$

$$\frac{1}{4} \int_{-1}^1 u^5 + u^4 du$$

$$\text{if } x = 1 \quad u = 1$$

$$x = 0 \quad u = -1$$

$$= \frac{1}{4} \left[\frac{u^6}{6} + \frac{u^5}{5} \right]_{-1}^1 = \frac{1}{4}$$

[3]

[2]

$$e) \alpha + \beta + \gamma = -2 \quad \alpha\beta + \alpha\gamma + \beta\gamma = -3 \quad \alpha\beta\gamma = -5$$

(-1 each error)

$$(\alpha-1)(\beta-1)(\gamma-1) = \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma)$$

$$= -5 - (-3) + (-2) = -1$$

$$= -5$$

[1]

(correct answer only here. No half marks)

QUESTION 2

$$a) \frac{d}{dx} [\ln(2x) - \ln(x-1)^2] = \frac{1}{2x} \cdot 2 - \frac{2}{x-1}$$

$$= \frac{1}{x} - \frac{2}{x-1}$$

[1]

(No part marks. correct or award 0) OR = $\frac{-x-1}{x(x-1)}$

$$b) \frac{dy}{dx} = \frac{1}{x^2 + 25}$$

$$y = \int \frac{1}{x^2 + 25} dx$$

$$= \frac{1}{5} \tan^{-1} \frac{x}{5} + c$$

$$\text{when } x = 5 \quad y = \frac{\pi}{2}$$

$$\frac{\pi}{2} = \frac{1}{5} \tan^{-1} 1 + c$$

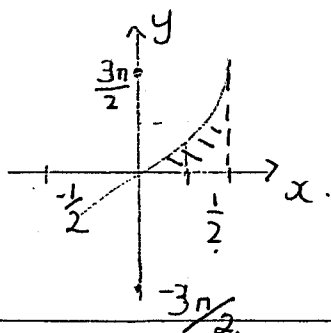
$$\frac{\pi}{2} = \frac{1}{5} \times \frac{\pi}{4} + c$$

$$\frac{9\pi}{20} = c$$

$$y = \frac{1}{5} \tan^{-1} \frac{x}{5} + \frac{9\pi}{20}$$

[2]

2c)



$$D: -1 \leq 2x \leq 1 \quad \therefore -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$R: -\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

[1]

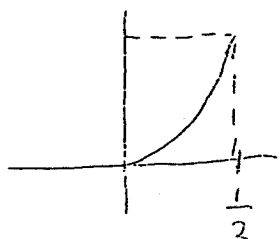
2e)

$$\begin{array}{r} 2x^2 - 7x + 15 \\ x^2 + 2x - 1 \overline{) 2x^4 - 3x^3 - x^2 + 2x + 1} \\ \underline{2x^4 + 4x^3 - 2x^2} \\ -7x^3 + x^2 + 2x \\ \underline{-7x^3 - 14x^2 + 7x} \\ 15x^2 - 5x + 1 \\ \underline{15x^2 + 30x - 15} \\ -35x + 16 \end{array}$$

$$Q(x) = 2x^2 - 7x + 15 \quad R(x) = -35x + 16$$

Award [2] if one error.

ii)



$$\int_0^{1/2} 3 \sin^{-1}(2x) dx \quad [3]$$

$$= \text{rectangle} - \frac{1}{2} \int_0^{3\pi/2} \sin \frac{y}{3} dy$$

$$y = 3 \sin^{-1}(2x)$$

$$\frac{y}{3} = \sin^{-1}(2x)$$

$$2x = \sin\left(\frac{y}{3}\right)$$

$$= \frac{1}{2} \times \frac{3\pi}{2} + \frac{3}{2} \left[\cos \frac{y}{3} \right]_0^{3\pi/2}$$

$$= \frac{3\pi}{4} + \frac{3}{2} \left[\cos \frac{\pi}{2} - \cos 0 \right]$$

$$= \frac{3}{4} (\pi - 2) \text{ units}$$

$$\text{or } \frac{3\pi}{4} - \frac{3}{2}$$

Award [1] for this
[-1 each error]

2d)

$$\begin{aligned} \int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}} &= \left[\sin^{-1} \frac{x}{\sqrt{3}} \right]_0^{\sqrt{3}} \\ &= \sin^{-1} \frac{\sqrt{3}}{\sqrt{3}} - \sin^{-1} 0 \\ &= \sin^{-1} 1 \\ &= \frac{\pi}{2} \end{aligned}$$

[2]

$$a) f(x) = \log_e 0.5 + \sin 0.5$$

$$= -0.214$$

$$f'(x) = \frac{1}{x} + \cos x$$

$$= \frac{1}{0.5} + \cos 0.5$$

$$= 2.8776$$

$$\text{approx} = 0.5 - \frac{0.214}{2.8776} = 0.574$$

$$b) \cos 2x = 2\cos^2 x - 1$$

$$\frac{1}{2} \int 1 + \cos 2\left(x - \frac{\pi}{4}\right) dx$$

$$\frac{1 + \cos 2x}{2} = \cos^2 x$$

$$\frac{1}{2} \left[x + \frac{1}{2} \sin 2\left(x - \frac{\pi}{4}\right) \right] + C$$

$$c) T_{k+1} = {}^{12}C_k (5x^2)^{12-k} \cdot \left(\frac{-1}{2x}\right)^k$$

$$= {}^{12}C_k \cdot 5^{12-k} \cdot x^{24-2k} \cdot (-1)^k \cdot \left(\frac{1}{2}\right)^k \cdot x^{-k}$$

$$\text{now } x^{24-2k-k} = x^9$$

$$24-3k=9$$

$$1k=5$$

$$\therefore \text{coefficient } {}^{12}C_5 \cdot 5^7 \cdot (-1)^5 \cdot \left(\frac{1}{2}\right)^5 [2]$$

$$= -1933503.75$$

$$d) \ddot{x} = -9x$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -9x$$

$$\frac{1}{2} v^2 = -\frac{9x^2}{2} + C$$

$$\text{when } x=4 \quad v=0$$

$$0 = -72 + C$$

$$C = 72$$

$$\frac{1}{2} v^2 = -\frac{9x^2}{2} + 72$$

$$\therefore v^2 = -9x^2 + 144$$

ii) max speed

acceleration = 0
at centre of motion
 $x=0$

$$v^2 = 144$$

$$v = 12 \text{ m/sec}$$

e) Join D to E

$\angle BED = 80^\circ$ $\angle$ in alternate segment

$\angle DOE = 80^\circ$ $2 \times$ angle at circumf.

$DO = OE$ (equal radii)

$\therefore \angle OED = 50^\circ$ (equal $\angle$ s isosceles Δ)

$$\angle BEO = \angle BED - \angle OED$$

$$= 80 - 50$$

$$= 30^\circ$$

QUESTION 4.

$$\begin{aligned} \frac{T_{k+1}}{T_k} &= \frac{{}^{12}C_k (2x)^{12-k} \cdot \left(\frac{3}{x}\right)^k}{{}^{12}C_{k-1} (2x)^{12-k} \cdot \left(\frac{3}{x}\right)^{k-1}} \\ &= \frac{{}^{12}C_k}{{}^{12}C_{k-1}} \cdot \frac{3}{2x} \\ &= \frac{12!}{(12-k)! k!} \cdot \frac{(12-k)! (k-1)!}{12!} \cdot \frac{3}{2x} \\ &= \frac{3(12-k)}{2k} \cdot \frac{1}{x} \\ &= \frac{39-3k}{2k} \end{aligned}$$

$$\frac{T_{k+1}}{T_k} \geq 1$$

$$\frac{39-3k}{2k} \geq 1$$

$$k \leq 7\frac{4}{5}$$

$$\text{so } k = 7.$$

greatest coeff

$${}^{12}C_7 \cdot 2^5 \cdot 3^7 [2]$$

$$= 55427328$$

[2]

b) i) $\sqrt{3} \cos 2t - \sin 2t = R \cos(2t + \alpha)$

$$R^2 = \sqrt{3}^2 + 1^2$$

$$R = 2$$

$$\frac{\sqrt{3}}{2} \cos 2t - \frac{1}{2} \sin 2t = \cos 2t \cos \alpha - \sin 2t \sin \alpha$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore 2 \cos\left(2t + \frac{\pi}{6}\right)$$

ii) $2 \cos\left(2t + \frac{\pi}{6}\right) = 0$

$$\cos\left(2t + \frac{\pi}{6}\right) = 0$$

$$2t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$2t = \frac{\pi}{3}, \frac{8\pi}{6}, \frac{14\pi}{6}$$

$$t = \frac{\pi}{6}, \frac{4\pi}{6}, \frac{7\pi}{6}, \dots$$

$$t = \left(\frac{3n+1}{6}\right)\pi$$

[2]

OR
 $2t + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{2}$

$$2t = 2n\pi \pm \frac{\pi}{2} - \frac{\pi}{6}$$

$$2t = 2n\pi + \frac{\pi}{2} - \frac{\pi}{6}$$

$$= 2n\pi + \frac{\pi}{3}$$

$$2t = 2n\pi + \frac{\pi}{2} - \frac{\pi}{6}$$

$$= 2n\pi - \frac{2\pi}{3}$$

$$t = \left(n\pi - \frac{\pi}{3}\right)$$

c) $x = 5 + \sqrt{3} \cos 2t - \sin 2t$ ①

$$\dot{x} = -2\sqrt{3} \sin 2t - 2 \cos 2t$$

$$\ddot{x} = -4\sqrt{3} \cos 2t + 4 \sin 2t$$

$$= -4(\sqrt{3} \cos 2t - \sin 2t)$$

$$= -4(x - 5) \text{ from ①}$$

[2]

ii) $x = 5 + 2 \cos\left(2t + \frac{\pi}{6}\right)$

centre is 5 amplitude 2

[1]

$\therefore$ moves between 3 and 7

iii) $x = 5 \quad 5 = 5 + 2 \cos\left(2t + \frac{\pi}{6}\right)$

$$2 \cos\left(2t + \frac{\pi}{6}\right) = 0$$

$$\therefore t = \frac{\pi}{6}$$

[1]

$$50) p = \frac{1}{3} \quad q = \frac{2}{3}$$

$${}^5C_3 p^3 q^2 = {}^5C_3 \left(\frac{1}{3}\right)^3 \cdot \left(\frac{2}{3}\right)^2 \quad \checkmark [1]$$

$$= \frac{40}{243}$$

$$ii) 1 - \text{no success} \geq 0.9$$

$$1 - {}^nC_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^n \geq 0.9$$

$$\frac{2}{3}^n \leq 0.1$$

$$\log_e \left(\frac{2}{3}\right)^n \leq \log_e 0.1$$

$$n \geq \frac{\log_e 0.1}{\log_e 2/3} \geq 5.4$$

[2]

at least 6 matches

$$5b) \begin{matrix} A & B \\ (-4, 1) & (2, 4) \end{matrix} \quad \begin{matrix} A & B \\ (-4, 1) & (2, 4) \end{matrix}$$

~~2:1~~ 2:1

$$Q = (0, 3) \quad R = (8, 7) \quad (2)$$

$$ii) PA = 2PB$$

$$\sqrt{(x+4)^2 + (y-1)^2} = 2 \sqrt{(x-2)^2 + (y-4)^2}$$

$$x^2 + 8x + 16 + y^2 - 2y + 1 = 4[x^2 - 4x + 4 + y^2 - 8y + 16]$$

$$\therefore 3x^2 - 24x + 3y^2 - 30y + 63 = 0$$

$$x^2 - 8x + y^2 - 10y = -21$$

$$(x-4)^2 + (y-5)^2 = 20$$

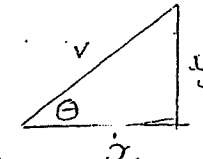
midpoint QR = (4, 5) which is

$$\text{length of QR} = \sqrt{8^2 + 4^2}$$

$$= \sqrt{80}$$

$$= 4\sqrt{5}$$

radius of circle = $2\sqrt{5}$ so QR diameter

$$c) \ddot{y} = -g$$


$$\dot{y} = -gt + c$$

$$\dot{y} = -gt + v \sin \theta$$

$$y = -\frac{1}{2}gt^2 + vt \sin \theta + c_2$$

$$\text{when } t=0 \quad y=0 \quad c_2=0$$

$$y = -\frac{1}{2}gt^2 + vt \sin \theta$$

$$\sin \theta = \frac{\dot{y}}{v} \quad \cos \theta = \frac{\dot{x}}{v}$$

$$v \sin \theta = \dot{y} \quad v \cos \theta = \dot{x}$$

$$\ddot{x} = 0$$

$$\dot{x} = v \cos \theta$$

$$x = vt \cos \theta + c_3$$

$$t=0 \quad x=0 \quad c_3=0$$

$$x = vt \cos \theta$$

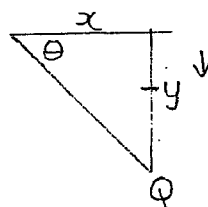
$$ii) y=0$$

$$0 = -\frac{1}{2}gt^2 + vt \sin \theta$$

$$0 = -gt + 2v \sin \theta$$

$$t = \frac{2v \sin \theta}{g} \quad [1]$$

$$iii)$$



$$\tan \theta = \frac{-y}{x}$$

$$= +\frac{1}{2}gt^2 - vt \sin \theta$$

$$vt \cos \theta$$

$$vt \cos \theta \cdot \tan \theta = +\frac{1}{2}gt^2 - vt \sin \theta$$

$$v \cos \theta \cdot \frac{\sin \theta}{\cos \theta} = +\frac{1}{2}gt + v \sin \theta$$

$$2v \sin \theta = \frac{1}{2}gt$$

$$\frac{4v \sin \theta}{g} = t$$

so twice the time to Q

QUESTION 6

a) $(1+x)^8 (1+x)^8 = (1+x)^{16}$

ie $({}^8C_0 + {}^8C_1 x + {}^8C_2 x^2 + \dots + {}^8C_8 x^8)({}^8C_0 + {}^8C_1 x + \dots + {}^8C_8 x^8)$

$= {}^{16}C_0 + {}^{16}C_1 x + \dots + {}^{16}C_{16} x^{16}$

[3]

RHS coeff of $x^8 = {}^{16}C_8$

LHS coeff of $x^8 = {}^8C_0 \cdot {}^8C_8 + {}^8C_1 \cdot {}^8C_7 + {}^8C_2 \cdot {}^8C_6 + \dots$

$= ({}^8C_0)^2 + ({}^8C_1)^2 + ({}^8C_2)^2 + \dots + ({}^8C_8)^2$

b) Prove true for $n=1$

LHS = $2 \times 1!$ RHS = $1(1+1)!$

$= 2$

$= 2$

$\therefore$ true for $n=1$

Step (2) Assume true for $n=k$

$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2+1)k! = k(k+1)!$

Step (3) prove true for $n=k+1$ if true for $n=k$

$= 2 \times 1! + 5 \times 2! + \dots + (k^2+1)k! + ((k+1)^2+1)(k+1)!$

$= k(k+1)! + (k^2+2k+1+1)(k+1)!$

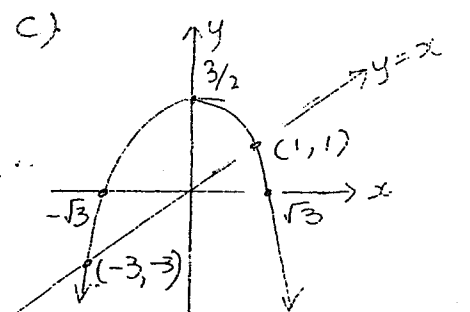
$= (k+1)! (k + k^2 + 2k + 2)$

$= (k+1)! (k+2)(k+1)$

$= (k+1)(k+2)!$

Step (4) since it is true for $n=1$ then it is true for $n=1+1, n=2$. Since

it is true for $n=k$ then it is true for $n=k+1$ and for all integral values of $n, n \geq 1$.

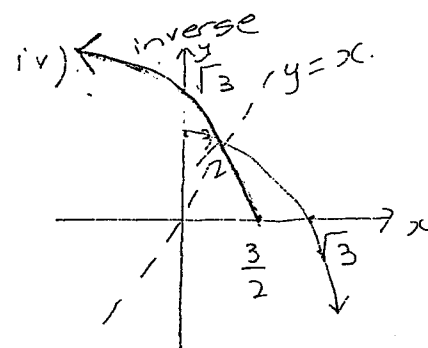


[2]

ii) $x \geq 0$ [1]

iii) D : $x \leq \frac{3}{2}$ [$\frac{1}{2}$]

R : $y \geq 0$ [$\frac{1}{2}$]



[1]

QUESTION 7.

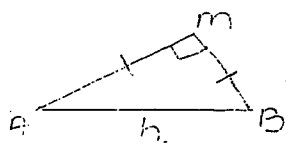
$$a) \cos \theta = \frac{AT^2 + AB^2 - TB^2}{2 \times AT \times AB}$$

$$= \frac{AT^2 + h^2 - AT^2}{2 \times AT \times AB}$$

since $AT = TB$

$$= \frac{h^2}{2 \times AT \times AB}$$

need to find AT^2 in terms of h .

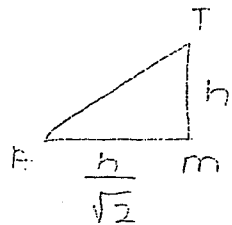


$AM = MB$ square pyramid

$$\therefore AM^2 + AM^2 = h^2$$

$$2AM^2 = h^2$$

$$AM = \frac{h}{\sqrt{2}}$$



$$AT^2 = h^2 + \frac{h^2}{2}$$

$$= \frac{3h^2}{2}$$

$$AT = h\sqrt{\frac{3}{2}}$$

[3]

$$\cos \theta = \frac{h^2}{2 \times h\sqrt{\frac{3}{2}} \times h}$$

$$= \frac{1}{2 \times \frac{\sqrt{3}}{\sqrt{2}}} = \frac{\sqrt{2}}{2\sqrt{3}} = \frac{1}{\sqrt{6}}$$

$$b) x = A \sin nt \quad T = \frac{2\pi}{n} \quad \text{so } 2\pi = \frac{2\pi}{n} \quad n=1$$

$$= A \sin t \quad [1]$$

$$ii) \tan \theta = \frac{x}{A}$$

$$A \tan \theta = x$$

$$\frac{dx}{d\theta} = A \sec^2 \theta$$

$$d\theta = A (1 + \tan^2 \theta)$$

$$= A \left(1 + \frac{x^2}{A^2} \right)$$

$$= A \left(1 + \frac{A^2 \sin^2 t}{A^2} \right) \quad [2]$$

$$= A (1 + \sin^2 t)$$

$$iii) \frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt}$$

$$= \frac{1}{A(1 + \sin^2 t)} \cdot A \cos t$$

$$= \frac{\cos t}{1 + \sin^2 t}$$

$$iv) \text{ Find when } \frac{\cos t}{1 + \sin^2 t} = \frac{2}{7}$$

$$7 \cos t = 2 + 2 \sin^2 t$$

$$7 \cos t = 2 + 2(1 - \cos^2 t)$$

$$2 \cos^2 t + 7 \cos t - 4 = 0$$

Q7 c+d.

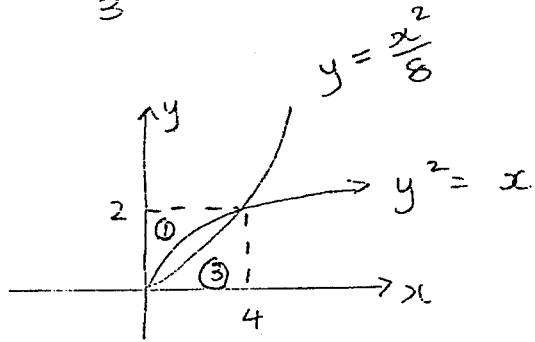
$$(2 \cos t - 1)(\cos t + 4) = 0$$

$$\cos t = \frac{1}{2} \quad \cos t = -4$$

no solution

$$t = \frac{\pi}{3}$$

(c)



$$x^2 = y^4 \quad \therefore 8y = x^2$$

$$8y = y^4$$

$$0 = y^4 - 8y$$

$$0 = y(y^3 - 8)$$

$$y = 0 \quad \text{or} \quad y = 2$$

$$x = 0 \quad \quad x = 4$$

$$\therefore a = 4 \quad b = 2$$

$$A_1 = \int_0^2 y^2 dy$$

$$= \frac{8}{3} u$$

$$A_3 = \int_0^4 \frac{x^2}{8} dx$$

$$= \frac{8}{3} u$$

$$A_2 = \text{rectangle} - A_1 - A_3$$

$$= 8 - \frac{8}{3} - \frac{8}{3}$$

$$= \frac{8}{3}$$

$\therefore$ rectangle divided in 3 equal areas